JALALUDEEN N

Senior Software Engineer

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PROFESSIONAL SUMMARY

ResuSenior Software Engineer with 6+ years of experience building high-scale, secure financial systems for global banking and fintech organizations. Hands-on experience developing Node.js microservices processing 10K+ daily requests, Open Banking integrations, and KYC onboarding platforms. Specialized in micro-frontend architectures using React.js, cloud-native AWS solutions, and CI/CD automation. Strong focus on API security, performance optimization, and regulatory-compliant delivery across European and Southeast Asian markets.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript, Java, HTML5, CSS3, C
Frontend Development: React.js, Next.js, Redux, Micro-Frontend Architecture, Component-Driven UI Design, Responsive Web Design, Jest, Playwright, Puppeteer
Backend Development: Node.js, RESTful APIs, GraphQL, Serverless Architecture, JWT, OAuth 2.0, API Design & Integration, Secure API Development
Cloud & DevOps: AWS (Lambda, S3, EC2, API Gateway, CDK), Docker, Jenkins, CircleCI, CI/CD Pipelines, Infrastructure as Code (IaC), Cloud-Native Application Development
Databases & Caching: MySQL, MongoDB, DynamoDB, Redis
Testing & Code Quality: Test-Driven Development (TDD), Unit Testing, Integration Testing, Jest, JUnit, Clean Code Practices, SOLID Principles, Design Patterns
Observability & Performance: Application Performance Optimization, Logging & Monitoring, Error Handling, Production Debugging
Tools & Platforms: GitHub, Git, VS Code, React Developer Tools, Chrome DevTools, Node.js CLI, npm, yarn, Postman, Swagger/OpenAPI, Serverless Framework, Jira
Architecture & Methodologies: Microservices Architecture, Event-Driven Systems, Agile (Scrum, Kanban), Secure & Scalable System Design, Performance Optimization
Operating Systems: Linux, macOS

PROFESSIONAL EXPERIENCE

Senior Software Engineer (Open Banking)
Client: American Express **May 2024 – Present**

- Key contributor to **American Express's first Open Banking implementation**, leading the rollout for the **France market** in compliance with Open Banking and regulatory standards.
- Designed and implemented **secure user redirection and return flows** involving external Open Banking providers (Algoan), ensuring seamless customer journeys without persisting sensitive user data in session or cache.
- Proposed and implemented a **state-resume mechanism** to restore user progress upon redirection return, improving customer experience while maintaining security and compliance.
- Developed and integrated **full-stack solutions** across **React.js (micro-frontend)** and **Node.js / Java backend services**, supporting high-availability banking workflows.
- Implemented **Server-Sent Events (SSE)** for real-time status updates, using **Redis Pub/Sub** to ensure consistent event delivery across multiple pods in a distributed Kubernetes environment.
- Designed **idempotent webhook processing** to handle duplicate and replayed Open Banking events, preventing multiple processing of the same customer consent or data payload.
- Handled complex **event-ordering scenarios**, where webhook data could arrive **before or after user redirection**, by implementing **Redis-based distributed locking mechanisms**.

- Ensured correctness in **cross-pod request handling**, where user requests and webhook callbacks were processed by different application instances.
- Implemented structured **application logging, correlation IDs, and dashboards** to enable real-time monitoring, root-cause analysis, and regulatory audit readiness.
- Led **production issue triage and hotfix deployment**, delivering same-day fixes to prevent large-scale transaction failures and customer impact.
- Contributed to **Java-based downstream services**, ensuring seamless interoperability across heterogeneous technology stacks.
- Implemented **Externalized Configuration Management (ECM)**, enabling dynamic configuration changes via centralized repositories without application redeployment, significantly improving operational agility and release safety.

Software Engineer – KYC & Open Banking
Client: Standard Chartered Bank — Jan 2020 – Apr 2024

- Designed and developed **enterprise-grade KYC verification platforms** to support secure and compliant customer onboarding across multiple geographies, including **Vietnam and Malaysia**, aligned with regional regulatory and data-residency requirements.
- Implemented **end-to-end digital onboarding workflows**, integrating identity verification, document validation, customer data enrichment, and risk checks within a highly regulated banking environment.
- Built and optimized **CI/CD pipelines** to automate build, test, and deployment processes, improving release reliability, reducing manual errors, and enabling faster time-to-market.
- Developed and secured **API integrations with third-party banks and external service providers**, ensuring compliant data exchange using industry-standard **authentication and authorization mechanisms**.
- Implemented **input validation, data integrity checks, and error-handling strategies** to ensure consistency and accuracy across KYC data flows.
- Optimized **batch-processing pipelines** handling large-scale KYC datasets, improving processing throughput, system performance, and operational scalability.
- Designed and implemented **resilient processing mechanisms** to handle partial failures, retries, and recovery in large-volume onboarding operations.
- Collaborated closely with **compliance, risk, legal, and operations teams** to ensure adherence to internal security policies, regulatory guidelines, and audit requirements.
- Implemented structured **logging, monitoring, and alerting mechanisms** to enhance system observability and support audits, investigations, and regulatory reporting.
- Participated in **production deployments, post-release monitoring, and incident management**, contributing to root-cause analysis and preventive system improvements.
- Supported **continuous platform enhancements**, balancing regulatory compliance, system stability, and evolving business requirements.

Application Developer
ThoughtWorks — Aug 2019 – Nov 2019

- Designed and implemented **reusable, modular components** to streamline recruitment and interview workflows, improving scalability and long-term maintainability.
- Applied **Test-Driven Development (TDD)** and industry-standard **design patterns** (Singleton, Factory) to deliver high-quality, testable, and maintainable code.
- Built an **end-to-end recruitment monitoring system** with real-time dashboards, increasing visibility into hiring pipelines and reducing manual tracking efforts.

EDUCATION

- Master of Computer Application - Coimbatore Institute of Technology, Coimbatore, April 2019
  Main coursework: Data Structures, Design and analysis of Algorithms, Computer Architecture, Artificial Intelligence, Database Systems, Operating Systems, Software Engineering.
- B.Sc Computer Science - Jamal Mohamed College, Trichy, April 2015